

Section A : Pure Mathematics [40 marks]

- 1 Find the series expansion of $\ln(\cos x)$ in ascending powers of x up to and including the term in x^6 . [3]

By putting $x = \frac{\pi}{3}$ in your result, show that $\ln 2 \approx \frac{\pi^2}{a} + \frac{\pi^4}{b} + \frac{\pi^6}{c}$, where a , b and c are positive integers to be determined. [2]

- 2 A drug company produces three types of flu vaccines: Releve, Xtraease and Influze. Each of these is a combination of three key ingredients: Formaldehyde, Thimerosal, and Neomycin. The table below shows the numbers of units of each of the three ingredients required in making batches of the three flu vaccines. It also shows the amounts of the ingredients available and the profit, in thousands of dollars on each batch of flu vaccines.

	Formaldehyde	Thimerosal	Neomycin	Profit in thousands of dollars (\$)
Releve	1	1	3	14
Xtraease	2	1	6	10
Influze	1	1	2	10
Amount available	60	k	140	

For instance, one batch of Releve requires one unit of Formaldehyde, one unit of Thimerosal and three units of Neomycin. The profit on a batch of Releve is \$14000 and there are 60 units of Formaldehyde available.

If all the ingredients available are to be used,

- (i) show that the number of batches of Influze made is independent of the number of units of Thimerosal available. [2]
- (ii) Find the range of k . [3]
- (iii) In the case where $k = 55$, find corresponding profit made if all the batches were sold. [2]
- 3 (a) The complex number z satisfies the relation $|z + 4 - i| = 2\sqrt{3}$. Describe in geometrical terms, with the aid of a sketch, the locus of the point C which represents z in an Argand diagram. [2]
- (b) The half line P has Cartesian equation $y = mx + 1, \quad x < 0$.
It is given that $m > 0$ and P is a tangent to C .
- (i) Sketch the half line P on the Argand diagram in (a). [1]
- (ii) Express the equation of the half line P in the form $\arg(z - \alpha) = \theta$ and determine the complex number which is represented by the point of contact of the half line P and the locus of point C . [4]
Deduce the exact value of m . [1]

- 4 Two planes p_1 and p_2 have vector equations

$$\mathbf{r} \cdot \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix} = 2 \quad \text{and} \quad \mathbf{r} \cdot \begin{pmatrix} 1 \\ 9 \\ -3 \end{pmatrix} = -6 \quad \text{respectively.}$$

- (a) (i) Find a vector which is parallel to both planes p_1 and p_2 . [2]

- (ii) Hence, find a Cartesian equation for the line of intersection, l_1 , of the two planes. [2]

- (b) The plane p_3 has equation $\mathbf{r} \cdot \begin{pmatrix} -1 \\ a \\ b \end{pmatrix} = 6$ where a and b are real numbers.

- (i) Find the values of a and b if the 3 planes p_1 , p_2 and p_3 intersect in the common line l_1 . [3]

- (ii) Find the relationship between a and b if the plane p_1 and the plane p_3 have the same perpendicular distance away from the origin. [2]

- 5 (a) It is given that $u_n = \frac{1}{n(n+1)(n+2)(n+3)}$, for $n \geq 1$, and that $v_n = (n+3)u_n$.

- (i) Show that $v_n - v_{n+1} = ku_n$, where k is an integer to be determined. [2]

- (ii) Deduce that $\sum_{n=1}^N u_n = \frac{1}{3} \left(\frac{1}{3!} - \frac{N!}{(N+3)!} \right)$. [3]

- (b) Prove, by induction or otherwise, that

$$\sum_{r=3}^n r3^{r+2} = \frac{1}{4} \left((2n-1)3^{n+3} - 729 \right), \quad n \in \mathbb{Z}^+, \quad n \geq 3. \quad [4]$$

- Hence, express $\sum_{r=1}^n r3^r$ in a similar form. [2]

Section B : Statistics [60 marks]

- 6** The student resource centre has 4 rooms which can be booked for students' activities in a day at a time. Requests of the booking of a room take place independently with a mean of 4 requests per day.
- (i) Find the probability that not all requests for the booking of a room can be met on any particular day. [2]
 - (ii) On any particular day during which more than 2 requests are received for the booking of a room, find the probability that all requests for the booking of a room can be met on that day. [3]
 - (iii) Find the probability that there are at most 3 days in a particular week (taking 1 week to be 5 schooling days) during which not all requests for the booking of a room for that day can be met. [2]
 - (iv) Find the least number of rooms that the student resource centre should have so that, on any particular day, the probability that a request for the booking of a room for that day has to be refused is less than 0.05. [3]
- 7** A shop has 30 ClearVision television sets and 35 FreePixel television sets for sales in a particular sale season. The probability that a particular set of ClearVisison can be sold in the sale season is 0.4 and the probability that a particular set of FreePixel can be sold in the sale season is 0.5. It may be assumed that the number of ClearVision television sets being sold is independent of the number of FreePixel television sets being sold.
- The shop makes a profit of \$300 for each ClearVision set sold and \$200 for each FreePixel set sold.
- (a) Let T denotes the total profit made by the shop in the sale season. Show that the value of $E(T)$ and $\text{Var}(T)$ are 7100 and 998000 respectively. [3]
 - (b) By using a suitable approximation,
 - (i) find the probability that the total profit made by the shop in the sale season is between \$7000 and \$8000. [2]
 - (ii) find the probability that the difference in profit made by the sales of ClearVision and FreePixel television sets in the sale season is more than \$200. [5]
- 8** (a) During a banquet, 4 married couples (Mr and Mrs Chua, Mr and Mrs Ng, Mr and Mrs Sim and Mr and Mrs Oun) and the host were seated in a round table. Find the number of possible seating arrangements if
- (i) there is no restrictions, [1]
 - (ii) the seats are numbered and each couple are seated together with the host seated between the Sim and Chua family. [3]
- (b) Given the word "DISTRIBUTIONS", find the number of arrangements of these letters such that
- (i) all the 'I's and 'S's are separated from one another ; [2]
 - (ii) the letter 'TI' appear as ordered pairs in the word. [2]

- 9 A junior college of 1600 students is made up of male and female students who are either in JC1 or JC2 as follows :

	JC1 Students	JC2 Students
Male	420	380
Female	450	350

A sample of 50 students is to be taken from the population to investigate the number of hours spent on studying per day by a student from this Junior College.

Describe how you could take a stratified sample of 50 students from the school. [2]

The number of hours spent on studying per day by a student is denoted by x hours. The student council claimed that the average number of hours spent on studying per day by a student is more than 9 hours. The sample values for a random sample of 50 students are summarized by

$$\sum x = 438 \quad , \quad \sum (x - 9)^2 = 45.$$

- (i) Calculate the unbiased estimates of the population mean and variance. [3]
- (ii) Carry out an appropriate test at 5% level of significance to determine if the student council has overstated the average number of hours spent on studying per day by a student. [3]
- Explain what is meant by the phrase “5% level of significance” in this context. [1]

The student council made another claim that the average number of hours spent on studying per day by a student is μ_0 . Given that there is sufficient evidence at the 5% significance level that the mean number of hours spent on studying per day by a student is not μ_0 , find the range of possible values of μ_0 . [3]

- 10 Events A and B are such that $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cup B) = 0.5$.

- (i) Determine whether A and B are mutually exclusive. [2]
- (ii) Determine whether A and B are independent. [2]

In a competition, 2 teams (A and B) will play each other in the best of 3 games. That is, the first team to win 2 games will be the winner. In the first game, both teams have equal chance of winning. In subsequent games, the probability of team A winning team B given that team A won in the previous game is p and the probability of team A winning team B given that team A lost in the previous game is $\frac{1}{3}$.

- (iii) Illustrate the information with an appropriate tree diagram. [2]
- (iv) Find the value of p such that the team A has equal chances of winning and losing the competition. [2]

- 11 (a)** With the aid of suitable diagrams, describe the difference between the least squares linear regression line of y on x and that of x on y . [2]

State the point at which the estimated regression line of y on x and that of x on y will both pass through. [1]

The equation of the estimated least squares regression line of y on x for a set of bivariate data is $y = a + bx$ and the corresponding equation of x on y is $x = c + dy$. Show that the square of the linear (product moment) correlation coefficient is bd . [1]

- (b)** A study is conducted with a group of dieters to see if the number of grams (x) of fat consumes per day is related to cholesterol level (y). The results were as follows.

x	5	9	13	7	11	3
y	160	α	229	β	225	136

- (i)** Given that the estimated least squares regression line of y on x is $7y = 68x + 786$, and that the linear (product moment) correlation coefficient is 0.985. Show that the values of α and β are 205 and 185 respectively. [4]
- (ii)** Predict the value of x when $y = 210$ and comment on the result. [2]
- (iii)** Sketch a scatter diagram to illustrate all the 6 pairs of data and the regression line y on x . [2]

~ End of Paper ~